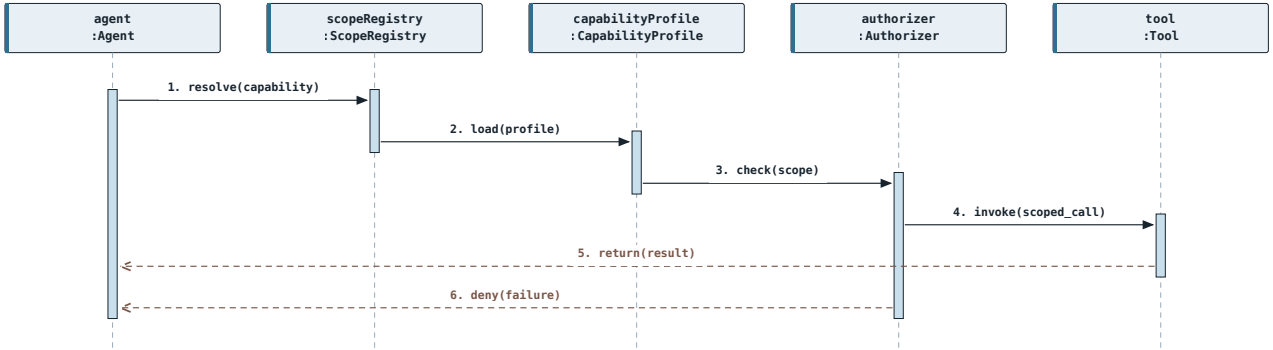
P25 Scope-based authorization and profiles

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Bind capabilities to explicit scopes.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A caller invokes a method or service with an operator, session, or agent identity.
TRIGGER	Capability is checked against a requested scope.
STRUCTURE	Scope identity, profile, grant and revocation state, and resolution function.
EFFECT	The call is permitted only if the principal currently holds a covering scope.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Principal identity and requested capability are explicit.
ENSURES	Authorization is evaluated at the call boundary and records the covering scope.
INVARIANT	Scopes are not inherited beyond declared containment or lifetime.
VIOLATION	Missing, expired, revoked, or cross-scope capabilities are denied explicitly.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.